Wall time per step: Parse 1M PBMC, full (684,932 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	69	34	OOM	OOM	40	OOM	OOM	731	OOM	OOM	142	384	143
qc	3.4	3.4	–	–	7.4	–	–	13	–	–	9.4	0.1	0.1
normalise	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.6	27	28
hvg	1.0	1.0	–	–	8.5	–	–	27	–	–	27	33	32
pca	4.4	4.1	–	–	17	–	–	106	–	–	696	38	41
harmony	3.7	3.4	–	–	4.5	–	–	255	–	–	217	228	240
knn_snn	14	13	–	–	33	–	–	61	–	–	70	167	172
cluster	6.4	5.9	–	–	30	–	–	217	–	–	256	17	17
cluster_fast	44	37	n/a	n/a	n/a	–	–	330	n/a	n/a	n/a	n/a	n/a
umap	50	50	–	–	349	–	–	262	–	–	454	588	588
dge	5.1	5.3	–	–	–	–	–	TIMEOUT	–	–	1,137	39	40
tsne_default	375	360	–	–	1,941	n/a	n/a	n/a	–	–	1,606	n/a	n/a
tsne_fft	113	112	–	–	247	n/a	n/a	n/a	–	–	232	n/a	n/a
tsne_quick	81	79	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	157	121	failed	failed	490	failed	failed	failed	failed	failed	3,012	1,521	1,300

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. Docker runs only; the native bixverse runs are in their own table.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.